



BioFuel Waste Tyre Pyrolysis Study

**Simulation, Economic, Environmental, and Digital Monitoring
Evaluation of a Small-Scale Waste Tyre Pyrolysis System**

BioFuel Company | Graduation Project
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Executive Summary



10–15	10	18–23	2,217	700,000
tons/day daily production	tons per reactor batch	hours/batch cycle time	JD/day expected net profit	JD estimated CAPEX

Core Finding
The proposed small-scale system is technically feasible, financially promising, and environmentally beneficial—provided that reactor reliability, product quality, and market access are managed carefully.

Main bottleneck
The pyrolysis reactor controls system throughput because it is batch-based and requires the longest processing time. Improving reactor cycle time or capacity would have the strongest positive impact on production, revenue, and waste conversion.

- Daily feed: 1000–1500 tyres, equal to 10–15 tons/day.
- The reactor is the common driver of throughput, profitability, and environmental benefit.
- Decision quality depends on real trials, quotations, market prices, permits, and downtime control.

Problem and Opportunity

The project addresses the gap between waste burden and resource recovery.

- End-of-life tyres are non-biodegradable and create environmental pressure.
- Pyrolysis converts tyres into oil, carbon black, gas, light fuel, and steel wires.
- A small-scale plant needs more than chemistry; it requires production-flow analysis.
- Discrete-event simulation helps reveal bottlenecks, waiting time, and utilization.

Research goal
Evaluate whether the proposed BioFuel system is technically, economically, and environmentally feasible.



Integrated Research Framework



Four connected analyses support one feasibility decision.

Simulation Model
Process flow
Capacity
Queues & utilization

Economic Feasibility
CAPEX/OPEX
Profitability
Payback & ROI

Environmental Assessment
Waste reduction
Energy recovery
Circular economy

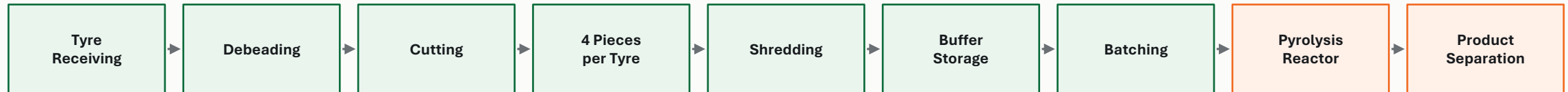
Digital Monitoring App
Machine status
Reactor tracking
Alerts & reports
Material levels
Temperature & humidity

Integrated conclusion
Reactor performance is the common driver of throughput, profit, environmental impact, and AI-based monitoring priorities.

Process Flow Definition



Waste tyres move through continuous preprocessing and batch-based thermal treatment.



Continuous preprocessing stages supply material to the buffer.

The reactor starts only after a full 10-ton batch is available.



General Production Data	Value
Average tyre weight	10 kg
Tyres per day	1000–1500
Daily production	10–15 tons/day
Pieces per tyre	4
Total pieces/day	4000–6000

Process	Processing Time	Unit	Capacity
Debeading	Triangular(0.2,0.3,0.5)	min/tyre	1
Cutting	Triangular(0.15,0.25,0.4)	min/tyre	1
Shredding	Triangular(0.25,0.4,0.6)	min/piece	1
Reactor	Triangular(18,20,23)	hours/batch	10 tons



BioFuel

BioFuel Company Data Report

Production and Operational Data for the Small-Scale Waste Tyre Pyrolysis System
Prepared as an official company data reference.

Table 1: General Production Data
This table summarizes the main production characteristics of the simulated system.

Parameter	Value
Average Tire Weight	10 kg
Tires per Day	1000–1500 Tires
Daily Production	10–15 tons/day
Pieces per Tire	4 Pieces
Total Pieces per Day	4000–6000 Pieces

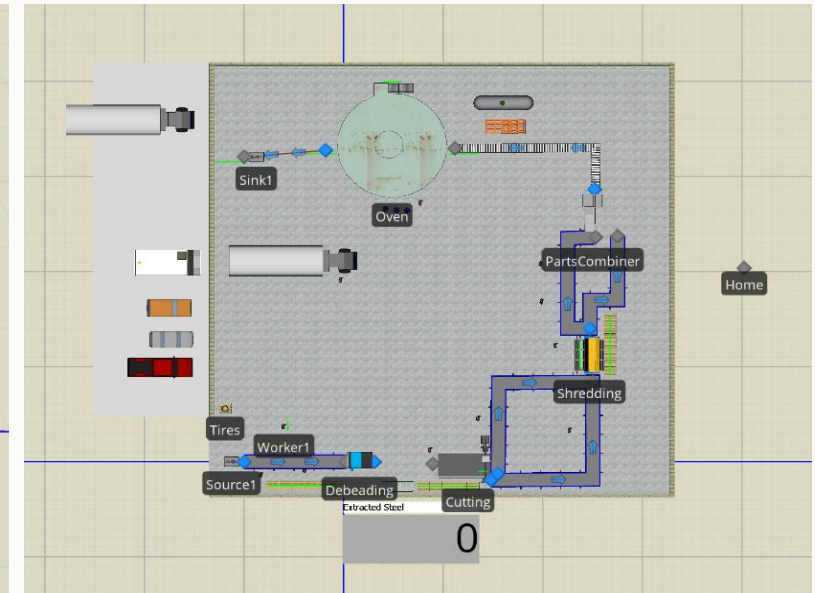
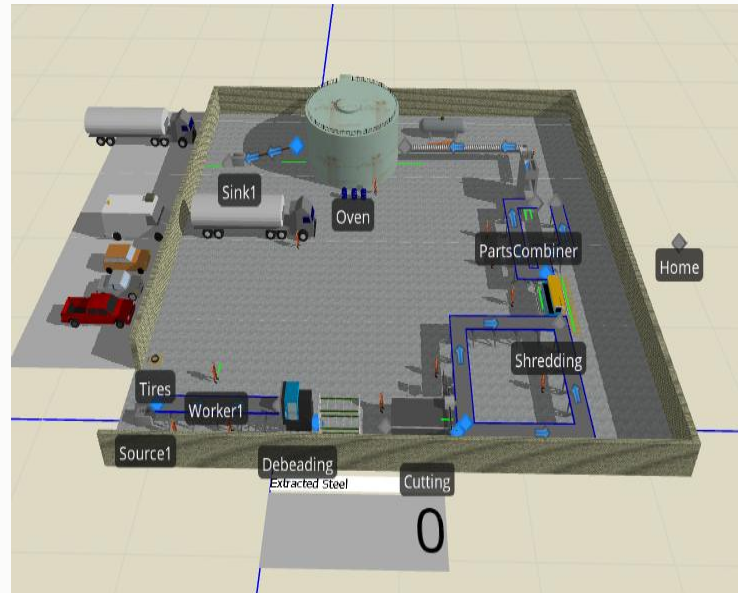
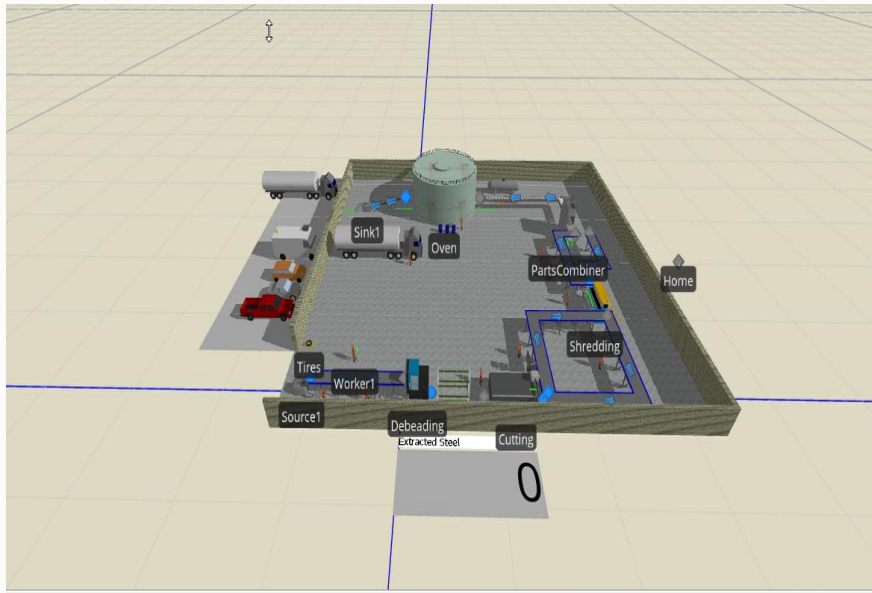
Table 2: Processing Times and Capacities
This table defines the processing times and capacity constraints of all system components.

Process	Processing Time	Unit	Capacity
Debeading	Random.Triangular(0.2, 0.3, 0.5)	min/tire	1
Cutting	Random.Triangular(0.15, 0.25, 0.4)	min/tire	1
Shredding	Random.Triangular(0.25, 0.4, 0.6)	min/piece	1
Buffer	—	—	Large
Reactor (Oven)	Random.Triangular(18, 20, 23)	hours/batch	10 tons

Seif Al-Mansarah
Seif Al-Mansarah
General Manager
BioFuel Company

Simulation System Model

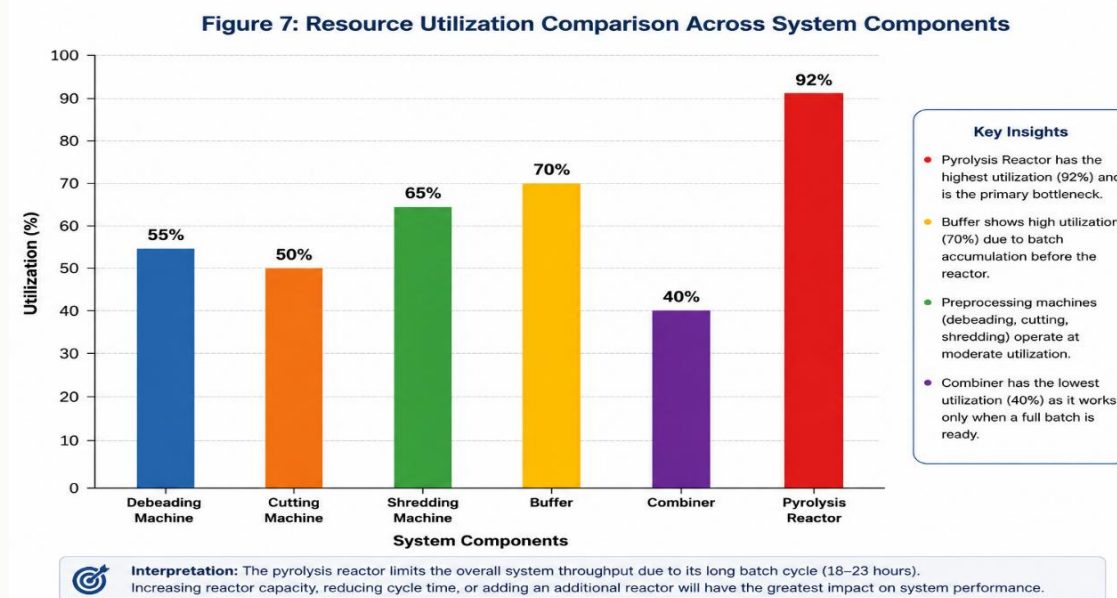
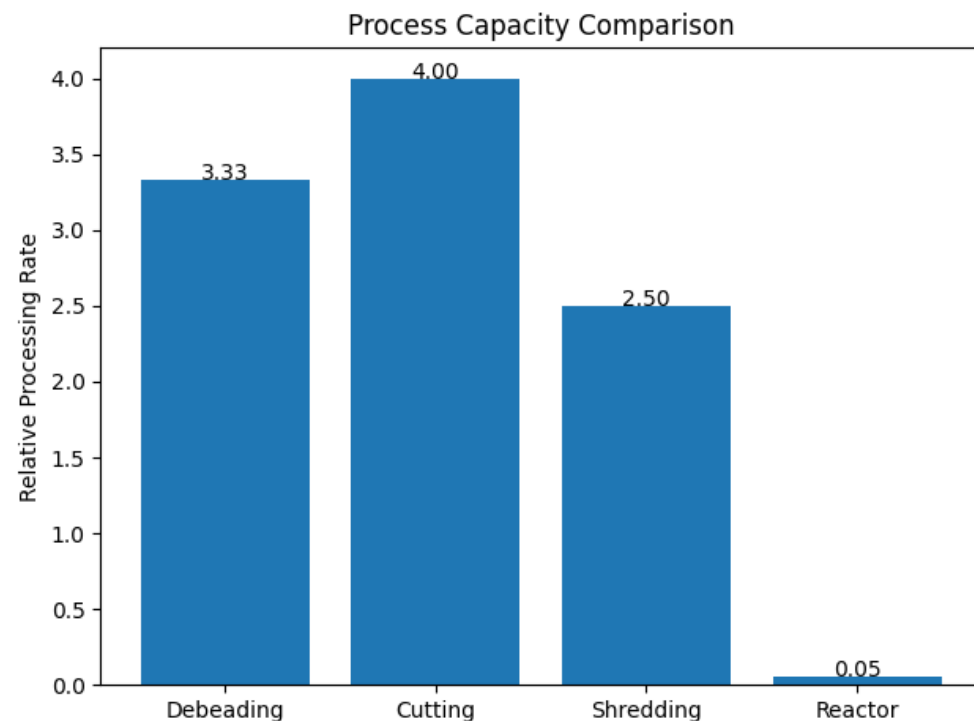
The system model links the physical plant flow with process capacity and queue behavior.



Model focus: receiving, preprocessing, buffer storage, batch formation, reactor processing, and product separation.

Simulation Results and System Performance

Simulation outputs show preprocessing is not the limiting area; reactor batching controls total performance.

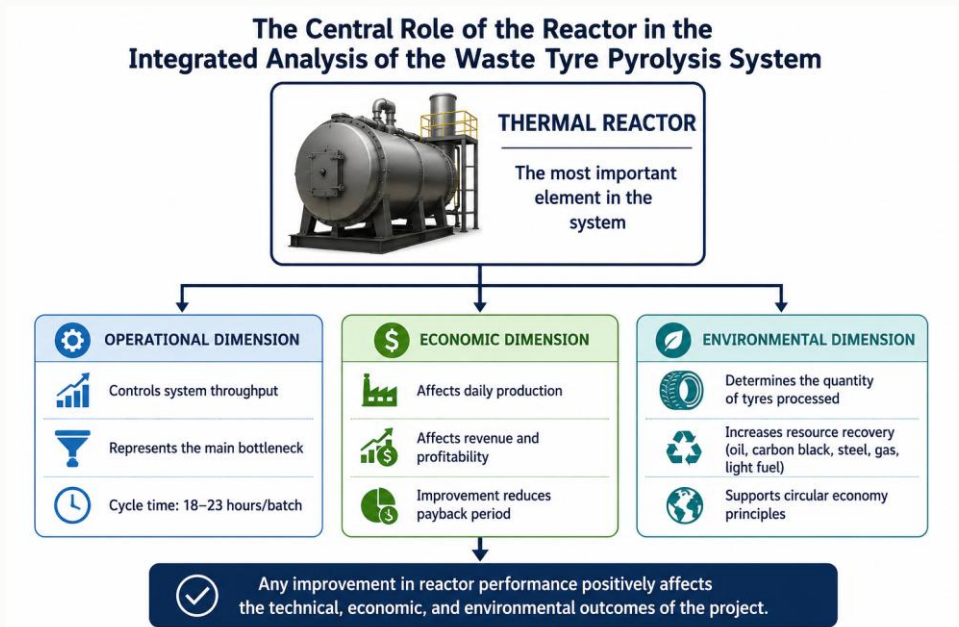


Key result: preprocessing has available capacity, while queues mainly form before batching and reactor loading.

Reactor Bottleneck and Utilization



The batch reactor controls total throughput and financial/environmental performance.



18–23	10
hours per batch	tons per batch
≈1	4000
batch per day	pieces per batch

Stage	Utilization
Debeading	38%
Cutting	40%
Shredding	55%
Buffer	70%
Reactor	94%

- Preprocessing stages have low waiting times.
- Material can accumulate enough for full reactor batches.
- Queues mainly form before batching and reactor.
- Reducing cycle time or adding reactor capacity is the direct path to higher throughput and lower waiting time.

Economic Feasibility — Investment and Revenue



Item	Estimated Cost (JD)
Pyrolysis reactor / oven system	250,000
Tire debanding, cutting, and shredding line	120,000
Condensation and oil recovery system	70,000
Carbon black and steel wire handling	40,000
Gas treatment and safety system	45,000
Electrical, piping, and utilities	60,000
Installation, commissioning, and transport	50,000
Civil works and storage area	40,000
Contingency and miscellaneous costs	25,000
Total Estimated CAPEX	700,000 JD

Product	Daily Profit Range (JD/day)
Diesel / pyrolysis oil	1,400–2,000
Carbon black	500
Light fuel fraction	100–150
Steel wires	150–200
Total Daily Product Profit	2,150–2,850

Financial caution
The final investment decision depends on market prices, product quality, environmental approval, and reactor downtime.

700,000 JD

estimated CAPEX

2,150–2,850

JD/day product profit

Economic Feasibility — Profitability and Payback

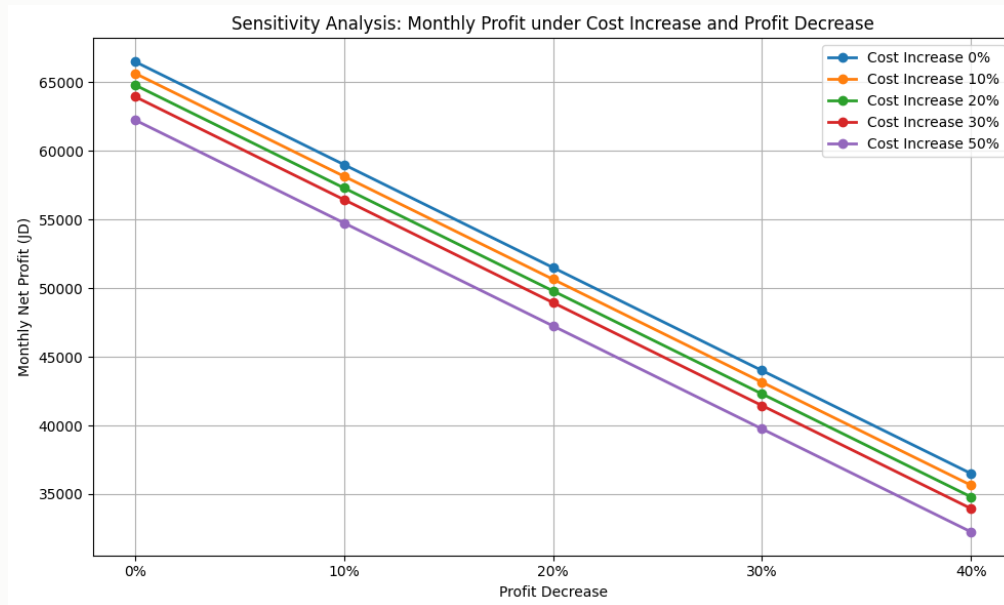


2,217	798,120	0.88	114%
JD/day expected net profit	JD/year expected net profit	years expected payback	expected ROI

Scenario	Daily Product Profit	Daily OPEX	Net Daily Profit	Monthly Net Profit	Annual Net Profit	Payback	ROI
Conservative	2,150 JD	333 JD	1,817 JD	54,510 JD	654,120 JD	1.07 yrs	93.45%
Expected	2,500 JD	283 JD	2,217 JD	66,510 JD	798,120 JD	0.88 yrs	114.02%
Optimistic	2,850 JD	233 JD	2,617 JD	78,510 JD	942,120 JD	0.74 yrs	134.59%

Scenario	Initial Investment	Annual Net Profit	Payback Period
Conservative	700,000 JD	654,120 JD	1.07 years
Expected	700,000 JD	798,120 JD	0.88 years
Optimistic	700,000 JD	942,120 JD	0.74 years

Sensitivity and Risk Drivers



Risk variable	Effect on feasibility
Pyrolysis oil price	Very high
Daily production rate	Very high
Reactor downtime	High
Product quality	High
Labor & energy cost	Medium

Sensitivity logic

The Python model tested cost increases and profit decreases across conservative, expected, and optimistic scenarios. Even under adverse changes, several scenarios remained feasible, but the margin depends strongly on product revenue and reactor uptime.

Most critical operational risk

Reactor downtime reduces throughput, product output, and revenue at the same time.

Environmental Impact and Resource Recovery

Recovered product	Yield
Pyrolysis oil	40%
Carbon black	30–40%
Gas	10%
Light fuel	10%
Steel wires	10%

$$\text{CO}_2 \text{ emission} = \text{Activity Data} \times \text{Emission Factor}$$

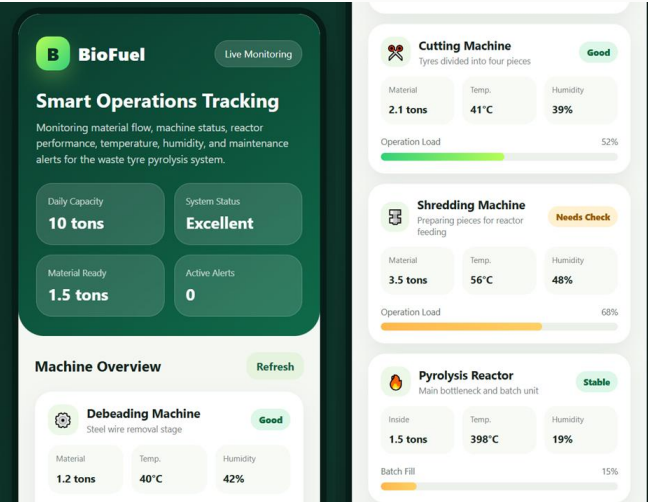


- Reduces tyre stockpiling and uncontrolled disposal.
- Recovers materials and energy from end-of-life tyres.
- Reuses gas internally as reactor heat source.

Digital Monitoring App and Integrated Discussion



The AI BioFuel mobile interface supports operational visibility through dashboard monitoring, reactor tracking, alerts, reports, material levels, temperature, and humidity.



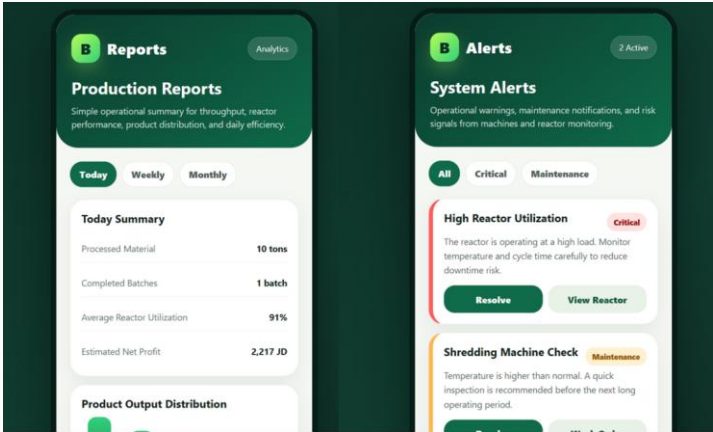
General Dashboard Interface

Technical
Controls throughput
and waiting time



Reactor-Specific Interface

Economic
Affects daily revenue
and payback period



Reports and Alerts Features

Environmental
Determines quantity
of tyres converted

Any reactor improvement produces a multi-dimensional benefit across the full feasibility model.

Recommendations and Conclusion



1

Improve reactor performance

Reduce cycle time, improve loading/unloading, and maintain high availability.

2

Secure product markets

Validate buyers for pyrolysis oil, carbon black, and steel wires before investment.

3

Strengthen quality control

Product quality affects selling price, compliance, and long-term market acceptance.

4

Plan environmental compliance

Include emission control, permits, safety systems, and gas treatment in project cost.

5

Use the BioFuel mobile app interface

Monitor plant condition through a simple dashboard.



Conclusion

The proposed BioFuel small-scale waste tyre pyrolysis system is technically feasible, economically promising, and environmentally beneficial under the stated assumptions.

Next step

Validate with real operating trials, supplier quotations, product quality testing, and environmental permitting.